Software Requirements Specification

for

Smart Tax Calculator

Experiment 3: Stakeholder Identification, Requirement Elicitation and User Stories

Software Engineering & Design Principles Lab (CS311L) — Semester V, Section D

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1. Introduction

1.1 Purpose of this Report

This report documents how the requirements of the Smart Tax Calculator were identified and gathered. It lists the stakeholders of the project, describes the elicitation techniques used (interviews, questionnaire, observation, user personas and document analysis), records what each technique revealed, and converts the findings into prioritised user stories. It builds on the Feasibility Report (Experiment 2) and feeds directly into the Software Requirements Specification (SRS) v1.0 and the UML models already prepared for the project.

1.2 Project Overview

The Smart Tax Calculator is a web-based application that estimates the income-tax liability of an individual taxpayer in India. The user enters income, deductions and exemptions, selects the Old or New Tax Regime (or both), and receives a slab-wise breakdown, a regime comparison and a plain-language summary. Administrators maintain slab and deduction reference data. The system is an academic project built with HTML/CSS/JavaScript, Java Spring Boot and MySQL, and its output is an estimate only, not professional tax advice.

1.3 Problem Being Addressed

Individual taxpayers currently calculate tax by hand, by spreadsheet, or with generic online calculators. These approaches are error-prone, hide how the figure was derived, and make it hard to decide between the two regimes. The elicitation activities below were designed to understand these pain points from the point of view of the people who face them.

2. Stakeholder Identification

A stakeholder is any person or group who uses, affects, is affected by, or has an interest in the system. Stakeholders were identified by analysing the problem statement, the use case diagram (actors: Taxpayer and Admin), the SRS user classes and the project environment.

2.1 Stakeholder Register

ID	Stakeholder	Type	Role / Relationship to the System	Main Interest and Concerns
S1	Salaried Employee	Primary (end user)	Enters salary components, exemptions and deductions; compares regimes during investment declaration and filing season	Quick, accurate result; clear regime recommendation; correct handling of standard deduction, HRA and 80C-type deductions
S2	Self-Employed / Professional	Primary (end user)	Enters business or professional income and eligible deductions	Support for non-salary income; simple input; trustworthy slab breakdown
S3	Individual Taxpayer (first-time / occasional)	Primary (end user)	Uses the tool once or twice a year with little tax knowledge	Plain language, guidance text, helpful validation, no jargon
S4	System Administrator	Primary (privileged user)	Updates slabs, rebate limits, cess, surcharge and deduction policies after each Union Budget	Easy rule updates without code change; secure, authenticated access; audit of changes
S5	Development Team (4 members)	Internal	Designs, codes, tests and documents the system	Clear, testable, feasible requirements within the semester
S6	Course Faculty / Lab Evaluators	Internal / sponsor	Assess completeness and quality of artefacts	Correct application of SE process; consistent documents
S7	Tax Professional / Chartered Accountant (domain expert)	Secondary	Validates tax rules and calculation logic	Accuracy of slab, rebate, surcharge and cess logic; visible disclaimer
S8	Testers / QA	Secondary	Derive test cases from requirements	Unambiguous, measurable requirements
S9	Documentation Writers	Secondary	Prepare user manual, FAQ and online help	Stable terminology and screen descriptions
S10	Income Tax Department (regulator)	Indirect / external	Publishes tax laws and slab rates; the system does not connect to it	Rules reflected correctly; product not presented as an official or filing tool
S11	Institution (KIET) / Hosting Environment	Indirect	Provides infrastructure for the final demonstration	Resource usage, compatibility, no special configuration

2.2 Power–Interest Analysis

Category	Stakeholders	Engagement Strategy
High power, high interest (manage closely)	S6 Faculty, S5 Development Team, S4 Administrator	Regular reviews, frequent feedback, direct involvement in decisions
Low power, high interest (keep informed)	S1, S2, S3 End users; S7 Domain expert	Interviews, questionnaire, prototype feedback, validation of tax logic
High power, low interest (keep satisfied)	S10 Regulator rules, S11 Institution	Follow official rules; meet deployment constraints
Low power, low interest (monitor)	S8 Testers, S9 Documentation writers	Share SRS and use cases; update when requirements change

3. Elicitation Plan and Techniques

No single technique captures every kind of requirement, so a combination was used.

Technique	Target Stakeholders	Why It Was Chosen	Output
Structured interviews	S1, S2, S4, S7	Gives depth on workflows, tax-specific rules and pain points	Interview summaries (Section 4)
Questionnaire (survey)	S1, S2, S3	Reaches many users quickly and gives quantitative evidence for prioritisation	Survey results (Section 5)
Observation	S1, S3	Shows what users actually do, including workarounds they do not mention	Observation notes (Section 6)
User personas	S1, S2, S3, S4	Turns findings into representative users that guide design decisions	Personas (Section 7)
Document analysis	Existing artefacts	Reuses knowledge from the Feasibility Report, SRS, UML class and use case diagrams, and official tax material	Consistency check and requirement baseline

Elicitation schedule

Activity	Participants	Duration
Stakeholder identification workshop	Project team	1 session
Interviews	4 interviewees	20–30 minutes each
Questionnaire (Google Forms)	30 respondents	3 days open
Observation sessions	3 participants	15 minutes each
Persona and user-story workshop	Project team	2 sessions

4. Interviews

Interviews were semi-structured. The same core questions were asked of every user, with additional questions for the domain expert and the administrator role.

4.1 Core Interview Questions

1. How do you currently estimate your income tax, and how long does it take?
2. What difficulties or mistakes have you experienced?
3. Which inputs (salary, other income, deductions) do you need to enter?
4. How do you decide between the Old and New Tax Regime today?
5. What would make you trust the result of an online calculator?
6. What would you want to see in the output (breakdown, summary, comparison, printout)?
7. Do you have any concerns about entering financial details online?

4.2 Interview Summaries

Interview 1 – Salaried Employee (S1) Uses a spreadsheet shared by the company's HR team and a few online calculators, which often give different answers. Finds it hard to know which deductions apply under the New Regime. Wants to enter salary, allowances, house-loan interest, insurance and investments once and see both regimes side by side. Does not want to give a name, PAN or any personal details. *Key needs:* regime comparison, automatic disabling of deductions not allowed under the new regime, no personal data, quick result.

Interview 2 – Self-Employed Freelancer (S2) Earns professional income plus some interest income. Is unsure how to combine different income heads. Wants a simple form and a clear total of gross income. Asked for a printable summary to share with an accountant. *Key needs:* inputs for business/professional and other income, printable summary, clear gross-income total.

Interview 3 – Chartered Accountant / Tax Professional (S7) Stressed that slab rates, rebate limits, surcharge thresholds and cess change with each budget, so the system must not hard-code them. Pointed out that cess applies after the rebate, deductions must be capped at statutory ceilings, and taxable income cannot be negative. Advised a visible disclaimer, and that the tool must not be used as the only basis for filing a return. Mentioned capital gains as a common but complex case that should be explicitly out of scope initially. *Key needs:* configurable rules, capped deductions, correct order of rebate and cess, disclaimer, explicit scope boundaries.

Interview 4 – Administrator / Project Maintainer (S4) Wants to update slabs and deduction limits through an admin screen or database tables, without recompiling the code. Wants each financial year stored separately so older years remain available. Asked that only authenticated administrators can change data. *Key needs:* admin maintenance of reference data per financial year and regime, authentication, separation of data from code.

5. Questionnaire

A 10-question online questionnaire was circulated to students, family members and working professionals. 30 responses were collected (illustrative sample for the academic project).

5.1 Questionnaire Items

1. What is your primary source of income? (Salary / Business or profession / Other)
2. How do you currently calculate your income tax? (Manually / Spreadsheet / Online calculator / CA or consultant)
3. How confident are you that you can choose the better tax regime? (1–5)
4. Which feature matters most? (Regime comparison / Slab-wise breakdown / Deduction guidance / Printable summary)
5. Would you be comfortable entering your name or PAN into a tax calculator? (Yes / No)
6. What is the maximum time you would wait for a result? (<3 s / 3–5 s / >5 s)
7. Which device will you mainly use? (Mobile / Laptop or desktop / Both)
8. Do you want to save past calculations? (Yes / No / Maybe)
9. Do you need an explanation of tax terms next to the fields? (Yes / No)
10. Any other suggestions? (open text)

5.2 Summary of Results

Question	Key Finding
Income source	53% salaried, 20% self-employed/professional, 27% other or occasional
Current method	40% use online calculators, 27% spreadsheets, 20% consult a CA, 13% calculate manually
Confidence in choosing a regime	Average 2.4 / 5 – most users are unsure
Most valued feature	Regime comparison (47%), slab-wise breakdown (27%), deduction guidance (17%), printable summary (9%)
Willingness to enter name/PAN	77% not comfortable
Acceptable wait time	83% expect a result within 3 seconds
Device	63% use mobile or both
Save past calculations	40% yes, 33% maybe, 27% no
Need explanation of terms	87% yes

5.3 Analysis

The survey confirms the interview findings: regime comparison is the most valued feature, users do not trust tools that ask for personal identity details, performance expectations are strict, and a majority need help understanding terminology. The high mobile usage justifies the responsive interface requirement. Interest in saving calculations is mixed, which matches the open item TBD-2 in the SRS (persisting calculation records).

6. Observation

Three participants were asked to compute their tax estimate using an existing online calculator while the team observed without intervening.

Observation	Implication for the System
Participants hesitated at deduction fields because section names (for example 80C, 80D) were unfamiliar	Add tooltips and plain-language labels next to every deduction field
One participant entered a deduction larger than the allowed limit and the tool silently accepted it, giving a wrong result	Cap deductions at the statutory ceiling and show a message stating the ceiling applied
A participant typed a comma-formatted number and got an error with a technical message	Accept sensible numeric formats and show validation messages in plain language, beside the field
Participants did not know why the final tax was a certain value	Show a slab-wise breakdown with rebate, surcharge and cess as separate lines
After switching regimes, one tool cleared all entered values	Preserve entered values when the regime is changed
On a phone, wide tables forced horizontal scrolling of the entire page	Use a responsive layout and let tables scroll within their own container

7. User Personas

Persona 1 – Aarav Sharma, Salaried Employee

Attribute	Detail
Age / Occupation	28, software engineer at a mid-size company
Tech skill	High
Tax knowledge	Moderate; knows 80C and HRA by name
Goals	Decide between Old and New Regime before submitting investment declaration; maximise take-home pay
Frustrations	Different calculators give different answers; unclear which deductions are allowed in which regime
Needs	Side-by-side regime comparison, automatic deduction handling, quick result on mobile
Quote	"Just tell me which regime saves me more, and show me why."

Persona 2 – Neha Gupta, Freelance Designer

Attribute	Detail
Age / Occupation	34, self-employed graphic designer
Tech skill	Moderate
Tax knowledge	Low to moderate
Goals	Estimate yearly tax to plan advance savings; share the summary with her accountant
Frustrations	Does not know how to combine professional and other income; worried about errors
Needs	Separate fields for business and other income, clear gross total, printable summary
Quote	"I don't want to hire someone just to know roughly what I owe."

Persona 3 – Ramesh Verma, Occasional Taxpayer

Attribute	Detail
Age / Occupation	52, government school teacher, uses a computer occasionally
Tech skill	Low
Tax knowledge	Low
Goals	Understand roughly how much tax applies to him and whether he gets a rebate
Frustrations	Technical terms, long forms, fear of making mistakes
Needs	Plain language, guidance text, large readable fields, clear error messages, a simple summary
Quote	"If it uses difficult words, I will just close it."

Persona 4 – Kavita Singh, Project Administrator

Attribute	Detail
Age / Occupation	26, developer who maintains the application
Tech skill	High
Goals	Update tax rules after each budget quickly and safely
Frustrations	Hard-coded values that need code changes and redeployment
Needs	Reference data in the database, per financial year and regime, restricted to authenticated administrators
Quote	"A budget change should be a data update, not a software release."

8. Consolidated Requirements

The elicitation outputs were merged, de-duplicated and mapped to the SRS requirement IDs.

8.1 Functional Requirements

ID	Requirement	Source	SRS Mapping
FR-1	Enter salary, allowances, house-property, business/ professional and other income	Interviews 1, 2; Survey Q1	REQ-1
FR-2	Accept only valid non-negative numbers; show plain-language errors next to the field	Observation; Interview 3	REQ-2, REQ-5
FR-3	Show running gross total income	Interview 2	REQ-4
FR-4	Enter deductions and exemptions with statutory ceilings shown	Interviews 1, 3	REQ-7, REQ-8
FR-5	Cap deductions at ceiling and notify the user	Observation; Interview 3	REQ-8, REQ-9
FR-6	Enable or disable deductions by regime	Interview 1	REQ-10
FR-7	Select Old, New or Compare; preserve data when switching	Interview 1; Observation	REQ-13, REQ-16
FR-8	Compute tax slab by slab with rebate, surcharge and cess; cess after rebate	Interview 3; Survey Q4	REQ-18 to REQ-22
FR-9	Compare both regimes, highlight the lower one and state the difference	Survey Q4 (47%); Interview 1	REQ-26 to REQ-29
FR-10	Generate plain-language tax summary and print-friendly view	Interview 2; Persona 3	REQ-31 to REQ-34
FR-11	Display a disclaimer on all result screens	Interview 3	REQ-30, REQ-36
FR-12	Admin maintains slabs, rebate, surcharge, cess and deduction policies per year and regime	Interview 4	Section 6.1; Business Rule 6
FR-13	Reset control to clear all values	Observation	REQ-6
FR-14	Optionally save a calculation without personal data	Survey Q8	Section 6.1 (TBD-2)

8.2 Non-Functional Requirements

ID	Category	Requirement	Source
NFR-1	Performance	Result within 3 seconds; comparison within 5 seconds	Survey Q6 (83%)
NFR-2	Privacy	No name, PAN or other personal identifiers required	Survey Q5 (77%); Interview 1
NFR-3	Usability	Tooltips and plain-language help; usable without tax background	Survey Q9 (87%); Persona 3
NFR-4	Portability / Responsiveness	Works on common browsers and on mobile screens	Survey Q7 (63%)
NFR-5	Correctness	Output matches manual calculation within one rupee; fixed-point decimals	Interview 3
NFR-6	Maintainability	Rules stored as configuration data, not in code	Interviews 3, 4
NFR-7	Security	Server-side validation, parameterised queries, authenticated admin access, no stack traces shown	Interview 4; SRS 5.3
NFR-8	Reliability	Preserve user data on error; refuse to compute if reference data is missing	Observation
NFR-9	Legal / Safety	Disclaimer cannot be removed; not a substitute for professional advice	Interview 3

8.3 Scope Boundaries Identified

- Capital gains income and other special-rate incomes are out of scope for Version 1.0 (SRS TBD-5).
- No connection to the Income Tax Department portal and no return filing.
- No login for ordinary taxpayers; authentication applies to administrators only.
- English interface only in Version 1.0, with externalised text for later translation.

9. User Stories

User stories follow the format: **As a** <role>, **I want** <goal>, **so that** <benefit>.

Epic A – Income and Deductions

ID	User Story	Acceptance Criteria	Priority
US-01	As a salaried employee, I want to enter my salary components and other income, so that my gross total income is calculated automatically.	Fields for basic salary, allowances, house property, business and other income; running total updates as values are entered; empty fields count as zero	Must
US-02	As a taxpayer, I want invalid entries to be rejected with a clear message, so that I can correct mistakes without confusion.	Negative, alphabetic and special-character entries are rejected; message appears next to the field in plain language; Next is blocked until errors are fixed	Must
US-03	As a salaried employee, I want to enter my eligible deductions and exemptions, so that my taxable income reflects what I am allowed to claim.	Standard deduction, investment, insurance and housing-loan interest fields available; each shows its ceiling	Must
US-04	As a taxpayer, I want amounts above the legal limit to be capped and explained, so that I do not accidentally overclaim.	Excess amount is ignored in calculation; message states the ceiling applied	Must
US-05	As a taxpayer, I want deductions that are not allowed in the New Regime to be disabled with a reason, so that I understand why they are not counted.	Unavailable categories are greyed out with an explanation when the New Regime is chosen	Must

Epic B – Regime Selection and Calculation

ID	User Story	Acceptance Criteria	Priority
US-06	As a taxpayer, I want to choose the Old Regime, New Regime or Compare, so that I can see the result I care about.	Exactly one option can be selected; Compare is the default; short description of the regimes is shown	Must
US-07	As a taxpayer, I want my entered values to be kept when I change the regime, so that I do not have to type everything again.	All income and deduction values retained; user is told if any deduction becomes inadmissible	Should
US-08	As a taxpayer, I want to see tax computed slab by slab, so that I understand how the total was reached.	Each slab shows range, rate, income in slab and tax; rebate, surcharge and cess shown as separate lines	Must
US-09	As a low-income taxpayer, I want to be told when my tax is nil and why, so that I know no tax is payable.	Nil liability message with reason when taxable income is within the exemption limit or rebate	Must
US-10	As a taxpayer, I want the result within a few seconds, so that using the tool is faster than doing it manually.	Single calculation under 3 seconds; comparison under 5 seconds	Must

Epic C – Comparison and Summary

ID	User Story	Acceptance Criteria	Priority
US-11	As a salaried employee, I want both regimes compared side by side, so that I can pick the one that costs me less.	Taxable income, total tax and effective rate shown for both; the lower-tax regime highlighted with the difference in rupees	Must
US-12	As a first-time taxpayer, I want a plain-language summary, so that I can understand it without tax knowledge.	Summary lists gross income, deductions, taxable income, tax, rebate, cess and total liability without unexplained section codes	Should
US-13	As a freelancer, I want a printable summary, so that I can share it with my accountant.	Print-friendly view through the browser; shows financial year, regime and disclaimer	Should
US-14	As a user, I want a visible disclaimer, so that I know the figures are estimates and not professional advice.	Disclaimer shown on every result screen and summary; cannot be removed	Must

Epic D – Administration and Data

ID	User Story	Acceptance Criteria	Priority
US-15	As an administrator, I want to update slabs, rebate limits, surcharge and cess for a financial year, so that the calculator stays current after each budget.	Reference data editable without code change; stored per financial year and regime	Must
US-16	As an administrator, I want only authenticated administrators to change rules, so that tax logic cannot be tampered with.	Rule-modification actions blocked for unauthenticated users	Must
US-17	As a taxpayer, I want the system to refuse to calculate if rules for the chosen year are missing, so that I never receive a result based on partial data.	Clear message shown; entered data retained	Should
US-18	As a privacy-conscious user, I want to use the calculator without giving my name or PAN, so that my identity stays private.	No personal identifiers requested for any calculation	Must
US-19	As a returning user, I want to optionally save a calculation, so that I can review it later.	Saving is explicit and stores inputs, result and timestamp only, with no	Could
US-20	As a mobile user, I want the screens to fit my phone, so that I can use the tool anywhere.	Layout is responsive; wide tables scroll inside their own container	Should

10. Requirement Prioritisation (MoSCoW)

Priority	User Stories	Rationale
Must have	US-01, 02, 03, 04, 05, 06, 08, 09, 10, 11, 14, 15, 16, 18	Core calculation, regime comparison, correctness, privacy and rule maintenance; required for the product to be useful and legally safe
Should have	US-07, 12, 13, 17, 20	Significantly improve usability and reliability but the product still works without them
Could have	US-19	Mixed survey interest (40% yes, 33% maybe); depends on resolving TBD-2
Won't have (this release)	Capital-gains handling, PDF export, multi-language UI, filing or portal	Outside the agreed scope and time frame (SRS TBD-4, TBD-5)

11. Conflicts, Risks and Assumptions

Item	Description	Resolution / Mitigation
Conflict	Users want to save calculations, but also do not want to share personal data	Saving is optional and stores no personally identifiable information
Conflict	Users want rich deduction support, but the New Regime allows very few deductions	Show only admissible deductions per regime and explain the disabled ones
Risk	Tax rules change every budget and the output may become outdated	Store rules as configurable data per financial year; display financial year and regime on every result
Risk	Users may treat the estimate as final advice	Permanent disclaimer; domain expert review of wording
Risk	Limited team time and experience	Prioritise Must-have stories; defer Could and Won't items
Assumption	Users enter accurate figures; the system does not verify them	Stated in SRS section 2.7
Assumption	The interview and survey sample is small and represents typical users, not all taxpayers	Validate again with a prototype demonstration

12. Traceability to Other Project Artefacts

Artefact	Link to Requirement Gathering
Use Case Diagram	Actors Taxpayer and Admin match stakeholders S1–S4; use cases (Enter Income, Enter Deductions, Select Regime, Calculate Tax, View Breakdown, Compare Regimes, Download Report, Manage Tax Slabs, Update Tax Rules) correspond to Epics A–D
UML Class Diagram	Classes TaxCalculationInput, Deduction, TaxRule, TaxSlab, DeductionPolicy, TaxCalculationResult, TaxComparison and TaxSummary realise FR-1 to FR-12
SRS v1.0	Requirements in Section 8 map to REQ-1 to REQ-36 and the non-functional sections; open items TBD-2, TBD-4 and TBD-5 are carried over
Feasibility Report	Technology stack, scope and constraints remain unchanged

13. Conclusion

The stakeholders of the Smart Tax Calculator were identified and classified, and their needs were gathered through interviews, a questionnaire, observation, personas and document analysis. The findings show that users most need a trustworthy regime comparison, a transparent slab-wise breakdown, help with unfamiliar terms, speed, and strong privacy, while administrators and the domain expert require configurable, correctly ordered tax rules and a visible disclaimer. These needs were consolidated into 14 functional and 9 non-functional requirements and expressed as 20 prioritised user stories with acceptance

criteria. The requirements are consistent with the SRS and UML models and form the basis for the next phases of design and implementation.

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